

PHYS 708 – Midterm (Fall 2025)

Conceptual questions (25 in total): Each carry 2 marks.

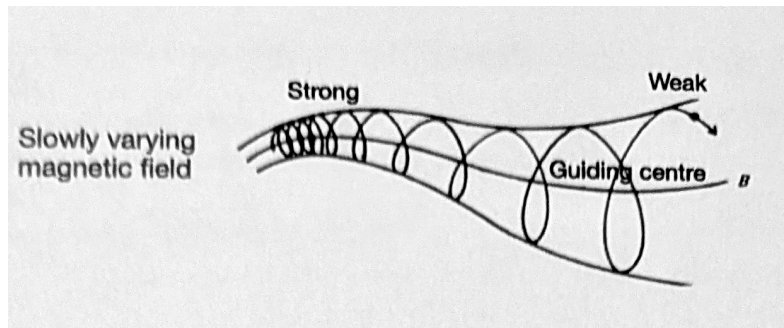
For multiple choice, circle your correct answer(s). For other type of questions, explain or prove your answer.

- (1) The energy densities of thermal matter, radiation photons, magnetic field and cosmic rays in the Galaxy are almost the same. Explain briefly the possible reason for such an equipartition in energy.

- (2) What kind of process(es) a high-energy positron can undergo when passing through a fully ionized medium?
 - (A) Ionization.
 - (B) Coulomb scattering.
 - (C) Bremsstrahlung.
 - (D) Annihilation.
 - (E) Inverse Compton scattering.

- (3) What is the primary source of the enormous amount of energy release during a core-collapse supernova explosion?
 - (A) Neutron degeneracy pressure.
 - (B) Electron degeneracy pressure.
 - (C) Gravitational potential energy.
 - (D) High temperature at the core of the star.
 - (E) Neutrinos generated at the core of the star.

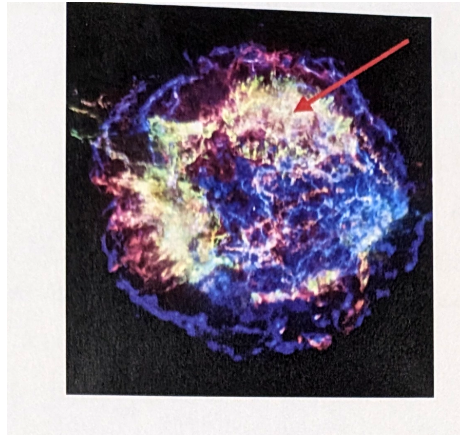
- (4) A high-mass star (e.g., having 15 solar mass) develops an onion shell structure with an iron core at the end of the nuclear fusion stage during its evolution. The iron core can then eventually end up into a neutron star that consists only of neutrons. Explain the process responsible for the creation of the neutrons inside the neutron star.



- (5) The figure above shows the trajectory of a charged particle in a slowly varying magnetic field. Choose the correct option(s) of the nature of the motion from the following.
- (A) The particle experiences an induced electric field which accelerates it along the field lines.
 - (B) The particle experiences an induced electric field which accelerates it in the radial direction.
 - (C) The particle experiences an induced electric field which accelerates it both along the field lines and in the radial direction.
 - (D) The particle does not get accelerated in any direction at all as magnetic fields cannot accelerate particles.
 - (E) The total kinetic energy of the particle remains constant.
- (6) The energy threshold of a cosmic-ray proton for $e^\pm$ pair production on Cosmic Microwave Background photon is 7×10^{17} eV. Choose the correct option for the following conclusion: In the rest frame of the proton, this energy *always* corresponds to a photon energy of ~ 1 MeV (2 times the rest mass of an electron).
- (A) True.
 - (B) False.
 - (C) Sometimes “true”, sometimes “false”.
 - (D) Can’t make a conclusion based on the given statement.
 - (E) None of the above.
- (7) For the interstellar medium, which of these processes is expected to dominate for cosmic-ray protons at energies below ~ 1 GeV/nucleon?
- (A) Escape from the Galaxy.
 - (B) Ionisation and elastic Coulomb scattering.
 - (C) Inelastic nuclear interactions with matter.
 - (D) Interaction with radiation fields.
 - (E) All of the above are equally important.
- (8) It is known that electron-positron annihilation at rest *always* produces two photons so that energy and momentum are conserved at the same time. Then, for the case of annihilation *in flight*, where one can assume that the positron is moving and the electron is at rest, can the process produce *one photon only* while conserving energy and momentum at the same time? Explain or prove through calculations.

- (9) Which high-energy processes are important for heavier nuclei, but not for protons in the Galaxy?
- (A) Ionization energy losses.
 - (B) Inelastic nuclear interactions with matter.
 - (C) Interaction with radiation fields producing electron-positron pair.
 - (D) Photopion production.
 - (E) Photo-disintegration.
- (10) Which component of the radiation field is distributed uniformly inside our Galaxy?
- (A) Infrared background.
 - (B) Optical starlight.
 - (C) Cosmic Microwave Background.
 - (D) Ultraviolet radiation.
 - (E) Diffuse gamma radiation.
- (11) For the same energy densities contained in thermal matter, ambient radiation photons and magnetic field, which of the following is (are) correct for the energy loss rate of highly energetic electrons?
- (A) Bremsstrahlung, Inverse Compton and synchrotron loss rates are all same.
 - (B) Bremsstrahlung and Inverse Compton loss rates are same.
 - (C) Bremsstrahlung and synchrotron loss rates are same.
 - (D) Inverse Compton and synchrotron loss rates are same.
 - (E) None of the loss rates are same.
- (12) If the observed highest energy cosmic rays are purely protons, the fast termination of the cosmic-ray spectrum beyond $\sim 10^{20}$ eV (the so-called GZK cut-off) can be due to the interaction of protons with
- (A) Matter producing pions.
 - (B) Cosmic microwave background (CMB) photons producing $e^\pm$ pairs.
 - (C) CMB photons producing pions.
 - (D) Thermal anti-protons undergoing annihilation.
 - (E) Magnetic field in the inter-galactic medium.
- (13) Which of the following mechanisms CANNOT accelerate charged particles?
- (A) Electric field.
 - (B) Uniform static magnetic field.
 - (C) Time-varying magnetic field.
 - (D) Random collisions with magnetized moving clouds.
 - (E) Shock waves running through an ionized medium.
- (14) First-order Fermi acceleration process is more efficient than second-order Fermi acceleration in accelerating cosmic rays. Explain why?

- (15) Which of the following options is (are) true for the diffusive shock acceleration process applied to supernova shocks?
- (A) The spectral shape depends on the mechanism of scattering on the upstream region.
 - (B) The spectral shape depends on the mechanism of scattering on the downstream region.
 - (C) The spectral shape depends on the mechanism of scattering on both the upstream and the downstream regions.
 - (D) The spectral shape depends on the shock compression ratio.
 - (E) The spectral shape depends on the size of the supernova.
- (16) Which process(es) is/are considered as continuous energy loss for energetic cosmic-ray protons?
- (A) Ionization and Coulomb collision losses with matter.
 - (B) $e^\pm$ pair creation on photon field.
 - (C) Photopion production.
 - (D) Deflection/scattering by magnetic field.
 - (E) Nuclear interaction with matter.
- (17) For efficient particle acceleration by the supernova shock waves, particles have to undergo diffusive motion on both sides of the shock. Which process is mainly responsible for producing the required magnetic scattering in the upstream region?
- (A) Magnetic fluctuation originally present in the interstellar medium.
 - (B) Alfvén waves generated by the streaming high-velocity charged particles.
 - (C) Magnetic fluctuation generated by the shock wave as it runs through the interstellar medium.
 - (D) The radiation energy released in the supernova explosion.
 - (E) The ejected material from the supernova explosion.
- (18) For a strong supernova shock producing a shock compression ratio of $r = 4$, the phase space particle index is 4.0. What is the phase space particle index for $r = 3$?
- (A) 3.5
 - (B) 4.0
 - (C) 4.5
 - (D) 5.0
 - (E) 5.5
- (19) The maximum energy of electrons accelerated by supernova shocks can be determined by which factor(s)?
- (A) Finite age of the shock.
 - (B) Free escape from the shock region.
 - (C) Synchrotron or inverse Compton energy loss.
 - (D) All of the above.
 - (E) None of the above (A, B, C).



- (20) The figure above shows a composite image of Cassiopeia A supernova remnant, where the bright colour (indicated by the arrow) represents X-ray emission. What is the origin of these X-rays?
- (A) Synchrotron radiation from high-energy electrons accelerated by the shock.
 - (B) Black-body (thermal) radiation from gas heated by the shock.
 - (C) X-rays from electron transition between the atomic energy levels.
 - (D) Bremsstrahlung emission from high-energy electrons.
 - (E) Inverse Compton emission from high-energy electrons.
- (21) Detection of high-energy TeV gamma rays from supernova remnants confirms
- (A) Acceleration of high-energy electrons.
 - (B) Acceleration of high-energy nuclei (mostly protons).
 - (C) Neither (A) nor (B).
 - (D) Both (A) and (B).
 - (E) Can't confirm either (A) or (B) with 100% certainty.
- (22) Detection of which high-energy messenger can be considered as the smoking gun for hadronic acceleration inside an astrophysical source? Explain why?
- (A) Cosmic-ray nuclei.
 - (B) Cosmic-ray electrons.
 - (C) High-energy X-rays.
 - (D) High-energy gamma rays.
 - (E) Neutrinos.

Write your reason here:

- (23) Which high-energy cosmic messenger given in the list below will be the best to study the most distant sources in the universe?
- (A) Cosmic-ray nuclei.
 - (B) Cosmic-ray electrons.
 - (C) High-energy gamma rays.
 - (D) Neutrinos.
 - (E) All of them.

Write your reason here:

- (24) The synchrotron energy loss rate for high-energy electrons follows $dE/dt \propto E^2$, where E is the energy of the electron. For such a loss rate, the energy loss timescale scales as
- (A) E .
 - (B) E^2 .
 - (C) $1/E$.
 - (D) $1/E^2$.
 - (E) some constant.
- (25) In the environment of an Active Galactic Nuclei, what is the dominant energy loss process for high-energy protons at the highest possible energies?
- (A) Ionization energy losses.
 - (B) Inelastic nuclear interactions with matter.
 - (C) Interaction with radiation fields producing electron-positron pair.
 - (D) Photopion production.
 - (E) Escape from the Galaxy.

Problem solving question (1 in total): 10 marks.

- (26) Show that the threshold kinetic energy of an incident proton to produce a neutral pion in a proton-proton interaction:

$$p + p \longrightarrow p + p + \pi^0$$

is given by

$$E_k = \frac{m_{\pi^0}}{2m_p} (m_{\pi^0} + 4m_p),$$

where m_{π^0} and m_p are the pion and proton rest mass energies.

You can use the next page to write your solutions.